

# A Packing of 14 Four-Spaces in $\mathbb{R}^{12}$

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## Abstract

We give 14 4-dimensional subspaces of  $\mathbb{R}^{12}$  whose smallest squared chordal distance is at least 2.871484. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.856823, obtained by deleting 1 of the 15 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 14 subspaces of dimension 4 in  $\mathbb{R}^{12}$  whose pairwise squared chordal distances  $D(a, b) = 4 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 2.871484 \dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 12$ ,  $n = 4$ . The cell  $N = 14$  is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.856823, obtained by deleting 1 of the 15 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is  $4(12 - 4)/12 \cdot 14/13 = 2.871795$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = YG^{-1}Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N14.txt`

in <https://github.com/Ge-limin/math719>.

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\*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 64 seconds on one laptop, with about ten such searches running at once.

## A Exact value

numerator:

```
287148415290911991892424900877466195287173519007144251498338925464171644
818705960328678298858457829684301192296273844598803664126290705317649375
7519302798098024360766543630378416998735279448031
```

denominator:

```
999999999999232628781344080861415559274086118769331417681296113702836514
192627622633983601790609728271625959897578739020478557395381412341233479
440449432804834676056273742861882301849588175120
```

## B Coordinates

14 blocks of 12 rows with 4 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m12n4N14.txt`.

```
-483712992458 152914346393 19492065395 -282825030159437450495 228182231475 543583686735 -28494862831024140763226 622027315078 159068237199 -214439467295
211699153401 350457506546 66954230196 -52108017716 381279390215 -260755286499 242944870099 -229227587482138106022 -542158323899 -152196882397 -60520119111
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